

Please write clearly in block capitals.

Centre number

--	--	--	--	--

Candidate number

--	--	--	--

Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL GCSE BIOLOGY

Paper 2

Tuesday 6 June 2023

07:00 GMT

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a ruler with millimetre measurements
- a scientific calculator.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- There are 90 marks available on this paper.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers.

Advice

In all calculations, show clearly how you work out your answer.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
TOTAL	



Answer **all** questions in the spaces provided.

0 1

The nervous system allows humans to react to changes in the environment.

0 1

. 1 Which cells detect changes in the environment?

[1 mark]

Tick (✓) **one** box.

Gland cells

☐

Muscle cells

☐

Receptor cells

☐

Stem cells

☐

A student investigated the effect of age on reaction time.

The student used an online program which measured the time taken to react to a change in colour of a computer screen.

This is the method used.

1. Sit person **A** facing a computer screen.
2. Tell person **A** to touch the screen as quickly as possible when the screen changes from red to green.
3. Record the reaction time displayed on the screen.
4. Repeat steps 2 and 3 four more times for person **A**.
5. Repeat steps 1 to 4 for person **B** and for person **C**.

0 1 . 2 Which variable should be a control variable in this investigation?

[1 mark]

Tick (✓) **one** box.

The age of the people

☐

The computer used

☐

The reaction time

☐

Question 1 continues on the next page

Turn over ►



Table 1 shows the results.

Table 1

Attempt number	Reaction time in milliseconds		
	Person A 50 years old	Person B 65 years old	Person C 80 years old
1	374	334	480
2	320	305	505
3	320	291	495
4	633	284	500
5	332	243	470
Mean	396	291	X

0 1 . 3 Calculate mean value **X**.

[2 marks]

Mean value **X** = _____ milliseconds

0 1 . 4 The mean for person **A** is incorrect.

What error has the student made when calculating the mean for person **A**?

Use information from **Table 1**.

[1 mark]



0 1 . 5

Each person did the test five times.

Which **two** are advantages of doing the test five times rather than just once?

[2 marks]

Tick (✓) **two** boxes.

To identify any anomalies

☐

To improve the reaction time

☐

To increase precision

☐

To increase repeatability

☐

To make sure it is a fair test

☐

0 1 . 6

The online program accurately measured the time taken for each person to react to a change in screen colour.

Why is it difficult to accurately measure reaction time using a stopwatch?

[1 mark]

Question 1 continues on the next page

Turn over ►



Many human actions are reflexes.

0 1 . 7 Give **one** feature of a reflex action.

[1 mark]

0 1 . 8 Which **two** are examples of reflex actions?

[2 marks]

Tick (✓) **two** boxes.

Blinking when an insect flies towards the eye

☐

Kicking a ball into a net during a game

☐

Putting on sunglasses in bright light

☐

Removing a hand from a hot object

☐

Writing an email to a friend on holiday

☐

0 1 . 9 Responses in reflex actions are carried out by effectors.

Which **two** are effectors?

[2 marks]

Tick (✓) **two** boxes.

Brain

☐

Gland

☐

Muscle

☐

Neurone

☐

Synapse

☐


0 2

One theory of evolution states that all life has descended from a universal common ancestor.

0 2 . 1

Complete the sentence.

Choose the answer from the box.

[1 mark]

complex

simple

specialised

The universal common ancestor was a _____ organism.

0 2 . 2

Complete the sentence.

Choose the answer from the box.

[1 mark]

adapted

extinct

separated

Over many generations changes within a species occur. A species not able to respond to changes in its environment may become _____.

Question 2 continues on the next page

Turn over ►



Table 2 shows a timeline for the evolution of the horse.

Table 2

Age of animal remains in millions of years	Name of animal	Height in metres	Bones in foot	Back teeth
1	Equus	1.60		
10	Pliohippus	1.25		
15	Merychippus	1.00		
35	Mesohippus	0.60		
50	Hyracotherium	0.40		

0 2 . 3

Which animal ancestor in **Table 2** did Mesohippus evolve from?

[1 mark]

0 2 . 4

Describe **two** differences between Pliohippus and Mesohippus.

[2 marks]

1 _____

2 _____



Figure 1

0 2 . 5 Explain why Mesohippus became extinct.

[6 marks]

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper has a slight shadow on the right side, suggesting it's resting on a surface.

Police horses have been conditioned to behave in specific ways.

0 2 . 6

Describe how conditioning can be used to train police horses to be calm in noisy environments.

[3 marks]

0 2 . 7

Police on horses are used to help manage large crowds of people.

Suggest **two** advantages of using police on horses compared to police on foot.

[2 marks]

1

2

0 2 . 8

Describe **one other** example where humans train an animal for a specific job.

[1 mark]



Turn over for the next question

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**

Turn over ►



0 3

Flowers are the reproductive organs in plants.

Female gametes and male gametes are produced in the flower.

0 3

1

Complete the sentence.

[1 mark]

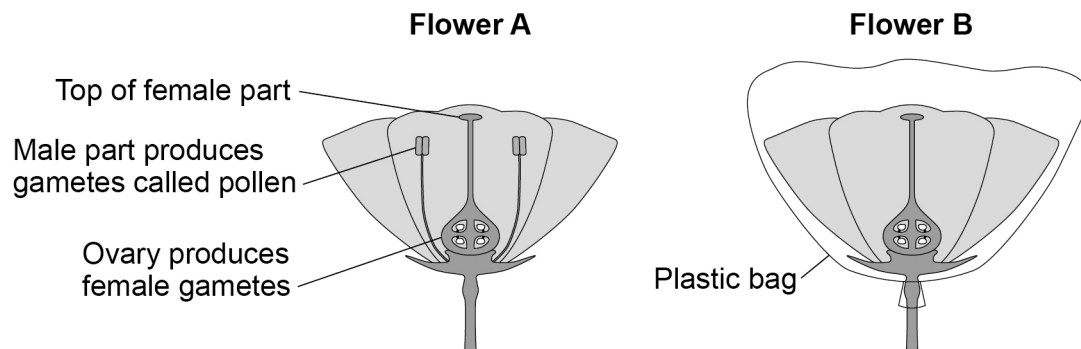
In sexual reproduction the female gamete and the male gamete _____.

New varieties of plants can be produced by a process called cross-pollination.

In cross-pollination the male gametes called pollen are transferred from one flower to the top of the female part of another flower.

Figure 2 shows two flowers.

Figure 2



A plant breeder placed pollen from flower **A** onto the top of the female part of flower **B**.



03.2

How did the plant breeder make sure that flower **B** was **only** cross-pollinated by flower **A**?

[1 mark]

03.3

Some plants can reproduce by self-pollination.

In self-pollination the female part receives pollen from the male part of the **same** flower.

How did the plant breeder make sure that flower **B** was **not** able to self-pollinate?

[1 mark]

Question 3 continues on the next page

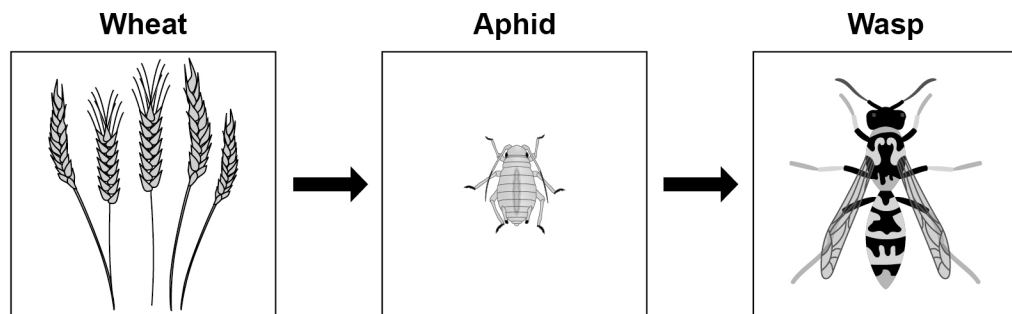
Turn over ►



Figure 3 shows a food chain in a farmer's field.

Aphids and wasps are insects.

Figure 3



Not to scale

0 3 . 4 Draw a pyramid of biomass for the food chain in **Figure 3**.

Label the pyramid.

[2 marks]

0 3 . 5

Suggest why farmers try to reduce the population of aphids in their fields.
Use **Figure 3**.

[1 mark]

Wheat can be genetically modified (GM wheat) to contain a gene from a peppermint plant.

The GM wheat produces a chemical which has a smell.

The smell acts as a warning signal for the aphids.

Aphids will stay away from the GM wheat.

0 3 . 6

Name the structure in a plant cell which contains genes.

[1 mark]

0 3 . 7

The required gene from the peppermint plant is isolated during genetic modification.

What is used to isolate the gene?

[1 mark]

Question 3 continues on the next page

Turn over ►

0 3 . 8 Farmers can reduce aphid populations by using GM wheat or by using pesticides.

Give **two** advantages of reducing aphid populations by using GM wheat compared to using pesticides.

[2 marks]

- 1 _____

2 _____

0 3 . 9 Give **two** reasons why some people are concerned about the use of GM wheat.

[2 marks]

- 1 _____

2 _____

12



0	4
---	---

Sanitiser is a type of liquid handwash.

A science class investigated the effect of two types of sanitiser on the number of bacteria left on the hands after washing.

Thirty students each prepared three agar plates.

The students labelled the agar plates:

- unwashed hands
- left hand washed with sanitiser **A**
- right hand washed with sanitiser **B**.

0	4	.	1
---	---	---	---

Some of the method is given below.

Give a reason for each of the following instructions.

[4 marks]

Heat the agar at 120 °C for 30 minutes.

Press fingertips onto the surface of the agar.

Secure the lid on the agar plate.

Incubate the agar plate for 48 hours.

Question 4 continues on the next page

Turn over ►



After 48 hours, the students counted the number of colonies of bacteria on each agar plate.

Table 3 shows the mean results from the thirty students.

Table 3

Sanitiser used	Mean number of colonies of bacteria
None (unwashed hands)	193
Sanitiser A	55
Sanitiser B	34

0 4 . 2

Calculate the percentage decrease in the mean number of colonies of bacteria for sanitiser **B** compared with unwashed hands.

[3 marks]

Percentage decrease = _____ %



The students carried out a second investigation to find out how effective the sanitisers were at preventing the growth of bacteria.

This is the method used.

1. Spread bacteria over the surface of an agar plate.
2. Take three paper discs:
 - soak one disc in sanitiser **A**
 - soak one disc in sanitiser **B**
 - soak one disc in distilled water.
3. Place the three paper discs on top of the agar.
4. Secure the lid on the agar plate and incubate for 48 hours.

0	4	.	3
---	---	---	---

Describe **three** techniques the students should use to prevent contamination when placing the paper discs onto the agar.

[3 marks]

1 _____

2 _____

3 _____

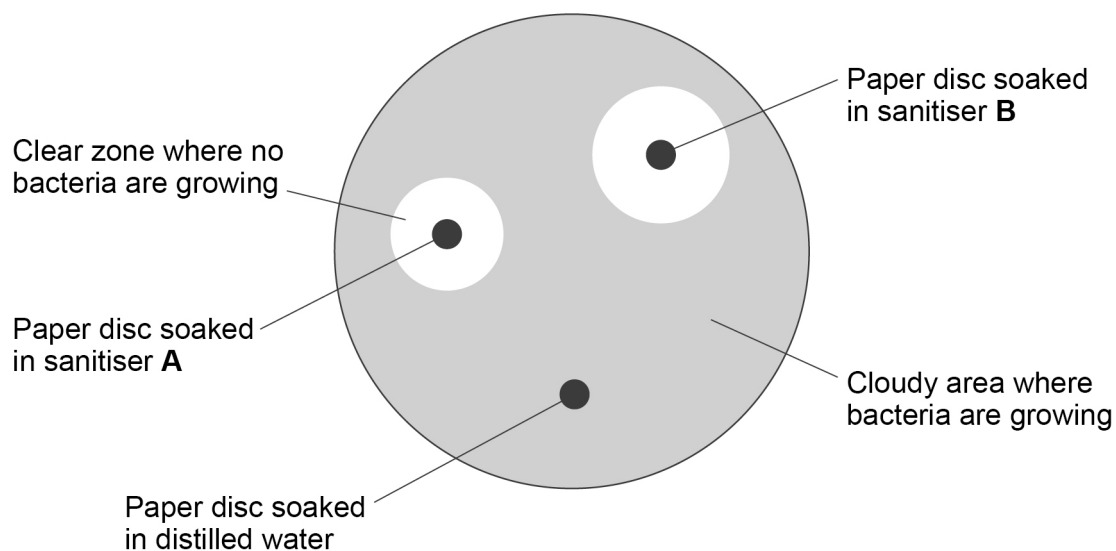
Question 4 continues on the next page

Turn over ►



Figure 4 shows the appearance of the agar plate after 48 hours.

Figure 4



0 4 . 4

The dependent variable in this investigation was the area of the clear zone.

Describe how the student can obtain a value for the area of a clear zone.

[3 marks]



0	4	.	5
---	---	---	---

Bacteria did not grow near the discs soaked in sanitiser but did grow at the edges of the clear zones.

Explain why.

[3 marks]

0	4	.	6
---	---	---	---

Sanitiser **B** contains a complex antimicrobial chemical.

Explain why widespread use of sanitiser **B** could increase the proportion of bacteria resistant to sanitiser **B**.

[2 marks]

18

Turn over for the next question

Turn over ►



0 5

Every year large areas of forest are cleared.

0 5 . 1

Explain how clearing large areas of forests affects the balance of gases in the atmosphere.

[2 marks]

The cleared land can be used to grow crops for fuel production.

Diesel and biofuel are two types of fuel that can be used in cars:

- diesel is made from fossil fuels
- biofuel is made from living plant material.

Table 4 shows information about diesel and biofuel when 1 kg of fuel is burned.

Table 4

	Diesel	Biofuel
Energy released in MJ	43	38
Carbon dioxide released in kg	3.17	2.52
Nitrogen oxide released in arbitrary units	100	110

Nitrogen oxide is a gas that has a similar effect on the environment as sulfur dioxide.

0 5 . 2

Describe **one** advantage of burning biofuel instead of burning diesel.

[1 mark]



0	5	.	3
---	---	---	---

Describe **two** disadvantages of burning biofuel instead of burning diesel.

Use data from **Table 4**.

[2 marks]

1 _____

2 _____

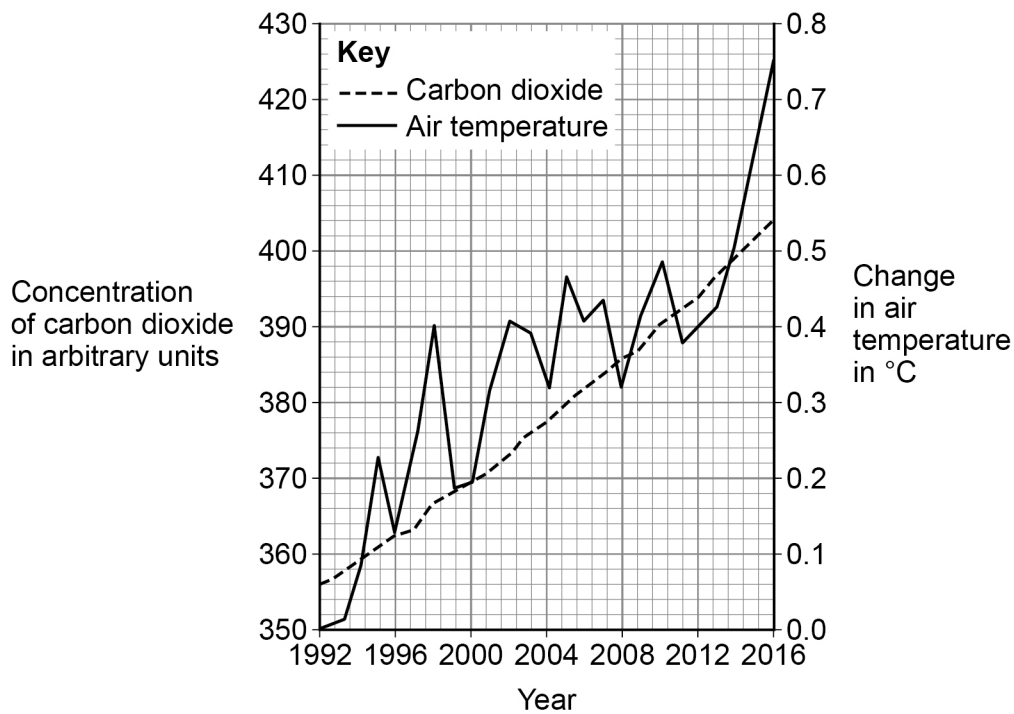
Question 5 continues on the next page

Turn over ►



Figure 5 shows information about the global atmosphere.

Figure 5



0 5 . 4

Many scientists think that the increase in air temperature is related to the increase in the concentration of carbon dioxide in the atmosphere.

Evaluate the evidence for and the evidence against the scientists' theory.

Use data from **Figure 5**.

[4 marks]



0	5	.	5
---	---	---	---

Explain how the increase in air temperature could lead to a decrease in the human population size.

[4 marks]

13

Turn over for the next question

Turn over ►



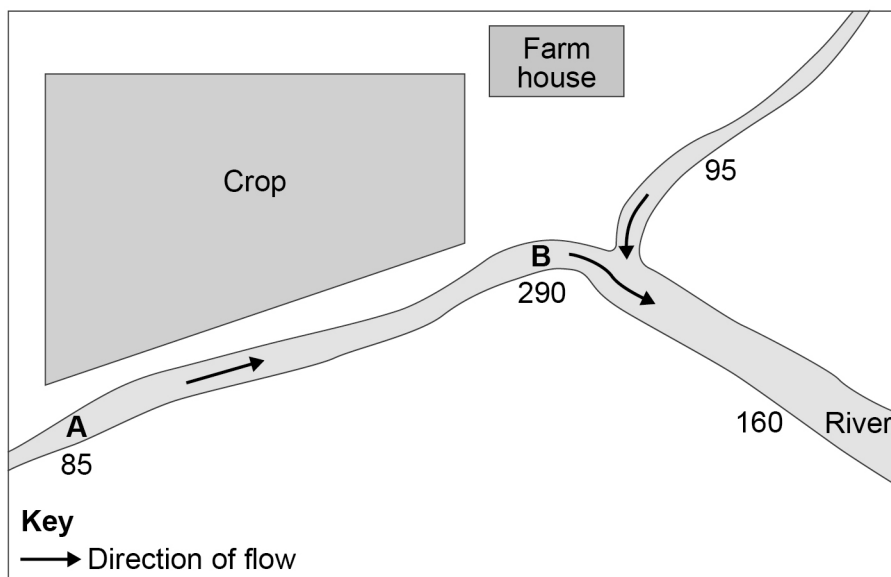
0 6

Farmers often use fertilisers on their crops.

Figure 6 shows a river system in an area where crops are grown.

The numbers show the concentration of nitrate ions in arbitrary units at different points in the river system.

Figure 6



0 6

. 1

The concentration of nitrate ions in the river is greater at **B** than at **A**.

Calculate how many times greater.

Give your answer to 2 significant figures.

[3 marks]

Number of times greater (2 significant figures) = _____



0 6 . 2Explain why the concentration of nitrate ions increased between **A** and **B**.**[2 marks]**

0 6 . 3

Explain how an increase in the nitrate ion concentration in the river may eventually lead to the death of fish.

[5 marks]

Weed plants in a crop affect the yield of the crop.

0 6 . 4

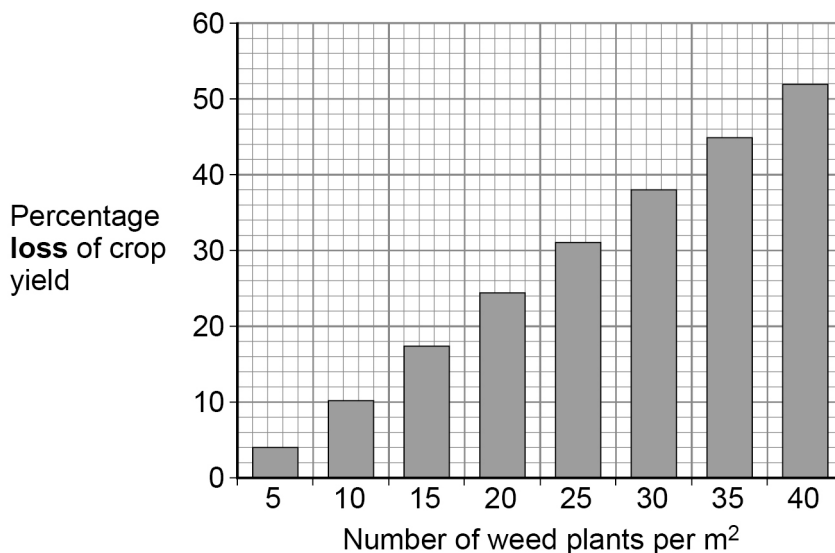
Explain why there is a higher crop yield if there are fewer weeds present.

[2 marks]

Turn over ►

0 6 . 5

Figure 7 shows the relationship between the number of weed plants per m^2 and the percentage **loss** of crop yield.

Figure 7

The farmer obtained a crop yield of 1.5 kg per m^2 when there were 5 weed plants per m^2 .

Calculate the crop yield when there were 40 weed plants per m^2 .

Use data from **Figure 7**.

[5 marks]

Yield = _____ kg per m^2

END OF QUESTIONS

There are no questions printed on this page

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**



[illegible]

Do not write
outside the
box

[illegible]

